CRISTY ZHANG

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EDUCATION

New York University, Tandon School of Engineering | B.S. Electrical and Computer Engineering May 2028 (Expected)

- Relevant Coursework: Computer Systems Architecture (Graduate), Computer Architecture, Embedded Systems, Digital Logic, Signals and Systems, Electronics, Object-Oriented Programming, Data Structures and Algorithms

SKILLS

HDL & Verification: SystemVerilog, Verilog, Icarus Verilog, GTKWave
Synthesis & Physical Design: Yosys, OpenROAD
Programming & Systems: C/C++, Python, Assembly (RISC-V), Linux
PCB Design & Simulation: Altium Designer, KiCad, LTspice
Embedded: STM32, Raspberry Pi, ESP32, Arduino, UART
Lab Equipment: Oscilloscope, Multimeter, Function Generator, DC Power Supply, Soldering

PROJECTS

32-Bit RISC-V CPU Design | GitHub: github.com/cristyz3077/32-bit-RV32I-CPU Aug 2026 - Present

- Designed a 32-bit RISC-V (RV32I) processor in SystemVerilog with a 5-stage pipeline (IF, ID, EX, MEM, WB), hazard detection, and data forwarding
- Developed modular CPU components, including ALU, register file, instruction decoder, immediate generator, control logic, program counter, and instruction/data memory
- Built self-checking SystemVerilog testbenches in Icarus Verilog and GTKWave, passing 81/81 custom integration checks across arithmetic, branch, memory, and jump instructions
- Running RTL synthesis and place-and-route with Yosys and OpenROAD to evaluate timing, area, and power

Autonomous Delivery Robot | GitHub: github.com/cristyz3077/autonomous-delivery-robot May 2026 - Present

- Developed STM32F446RE firmware implementing PWM motor control, encoder feedback, and PID-based speed regulation
- Established UART communication between STM32, Raspberry Pi, and PC for motor commands, telemetry, and debugging
- Designed a custom PCB in Altium Designer integrating the STM32, motor drivers, and power circuitry to replace the breadboard prototype

Sound Vest Jan 2025 - May 2025

- Designed a wearable vest that converts pre-recorded and live music into tactile vibrations, improving accessibility for people with hearing impairments
- Implemented real-time FFT audio processing on Raspberry Pi and Bluetooth communication between a mobile device, Raspberry Pi, and Arduino using HC-05 modules
- Developed Arduino firmware driving vibration motors with PWM signals mapped to detected audio frequencies and amplitudes

EXPERIENCE

System Administration Head, NYU General Engineering Department - New York, NY Sep 2025 - Present

- Collaborate with NYU IT to maintain and troubleshoot course computer systems, ensuring reliable lab operations
- Develop terminal scripts to automate computer troubleshooting
- Lead daily engineering labs on CAD, circuitry, and programming, mentoring 2–4 student project teams per semester and supporting students during Open Lab hours with project guidance, code debugging, and engineering concepts

Controls Subteam Member, NYU ARC Robotics: Team Ultraviolet - New York, NY Sep 2025 - Present

- Develop and debug STM32 firmware for ARC Robotics referee system integration, troubleshooting video transmission (VT) transmitter-receiver communication to enable accurate robot movement from keyboard and mouse commands
- Performed supercapacitor sizing calculations to determine optimal voltage ratings and capacitance for the robot's power system
- Simulated buck-boost converter circuits in LTspice and analyzed waveforms to verify expected circuit behavior

ACTIVITIES

Marketing Chair, Society of Asian Scientists and Engineers Sep 2025 - Present